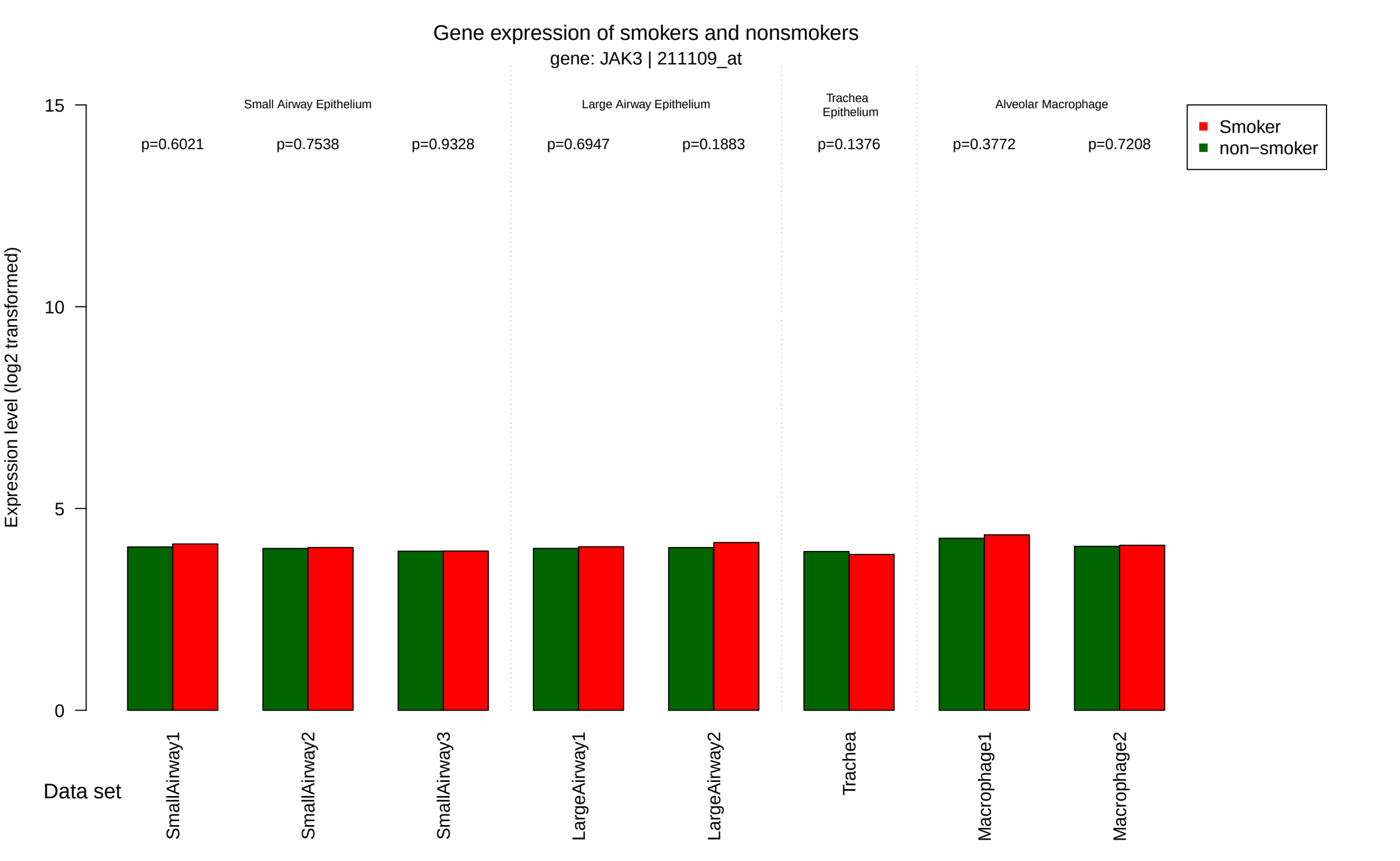
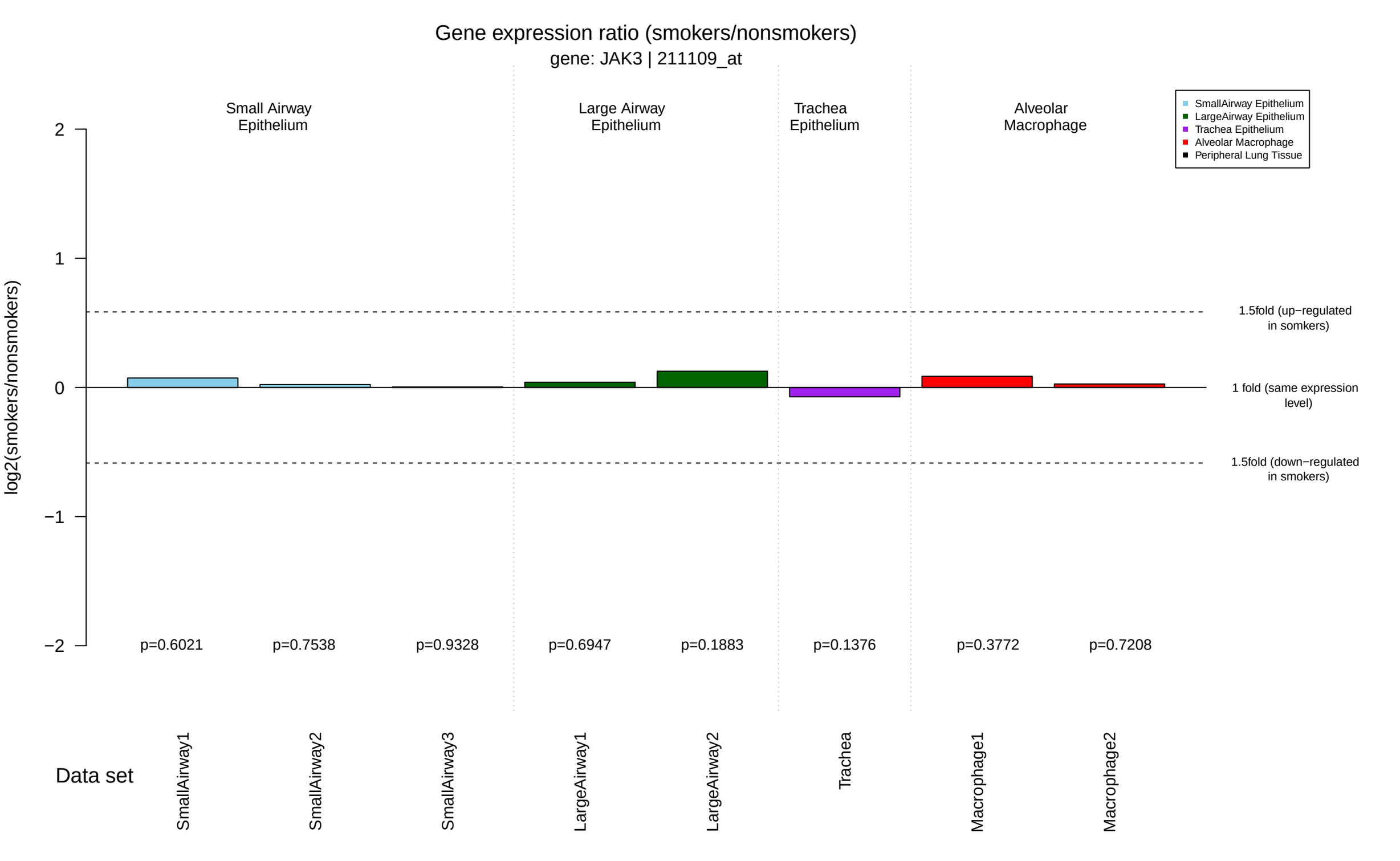
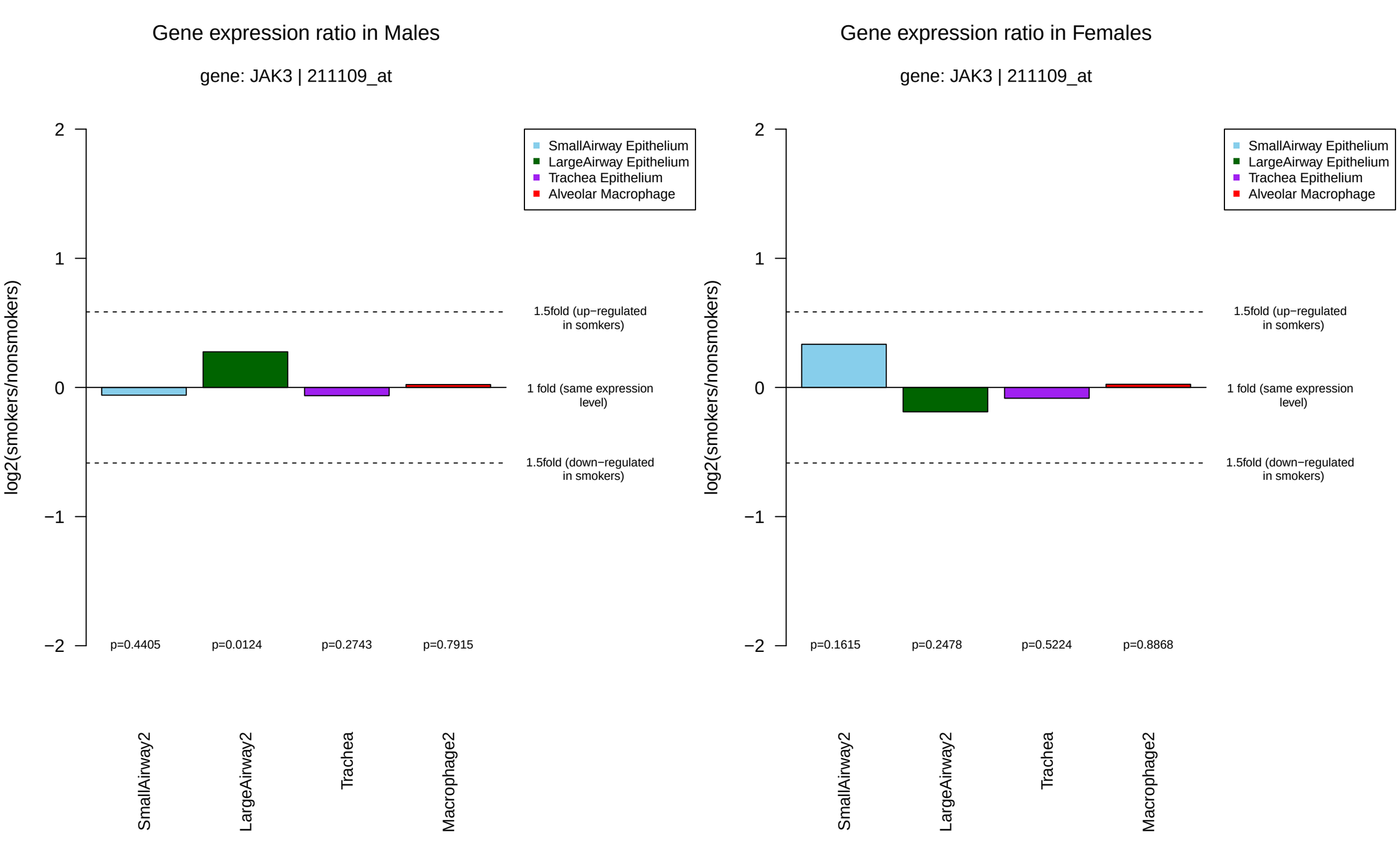
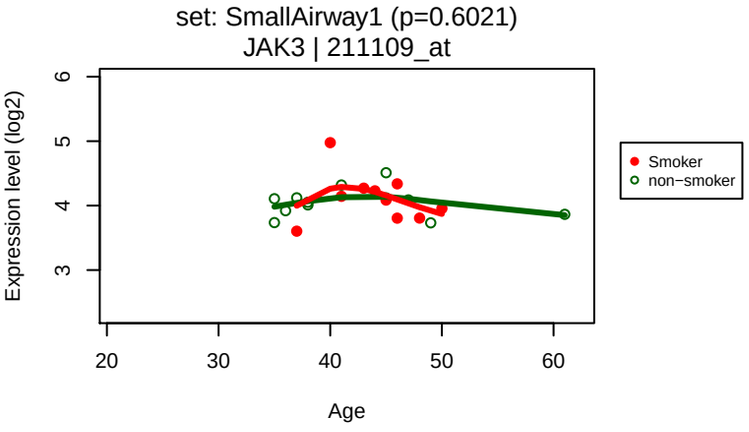
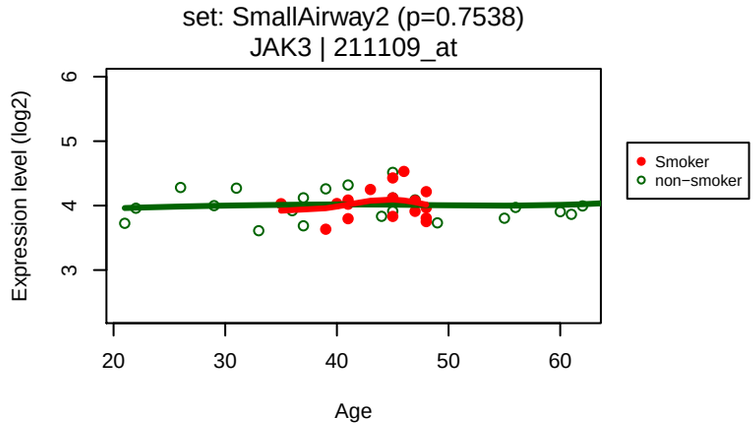
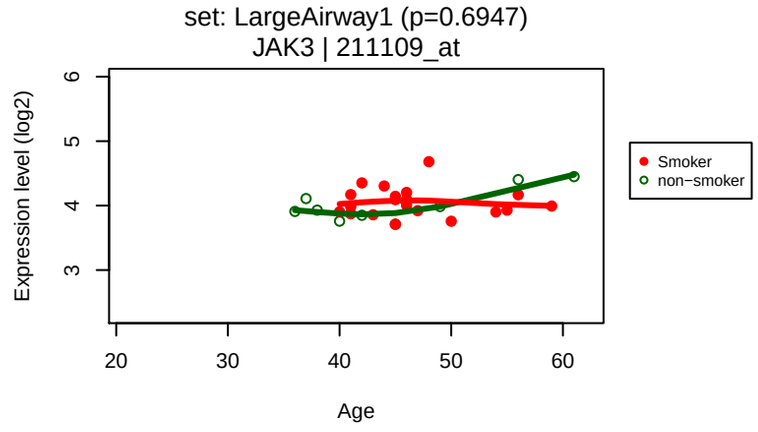

Gene expression of smokers and nonsmokers
gene: JAK3 | 211109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6021
p=0.7538
p=0.9328
p=0.6947
p=0.1883
p=0.1376
p=0.3772
p=0.7208
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

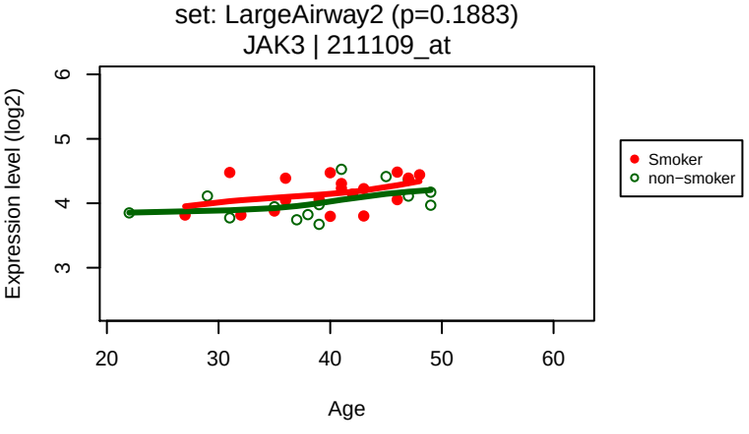

Gene expression ratio (smokers/nonsmokers)
gene: JAK3 | 211109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6021
p=0.7538
p=0.9328
p=0.6947
p=0.1883
p=0.1376
p=0.3772
p=0.7208
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

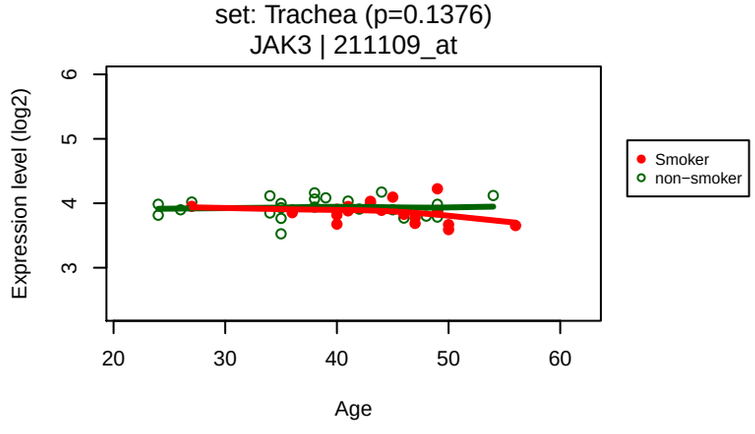

Gene expression ratio in Males
gene: JAK3 | 211109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4405
p=0.0124
p=0.2743
p=0.7915
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK3 | 211109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1615
p=0.2478
p=0.5224
p=0.8868
SmallAirway2
LargeAirway2
Trachea
Macrophage2

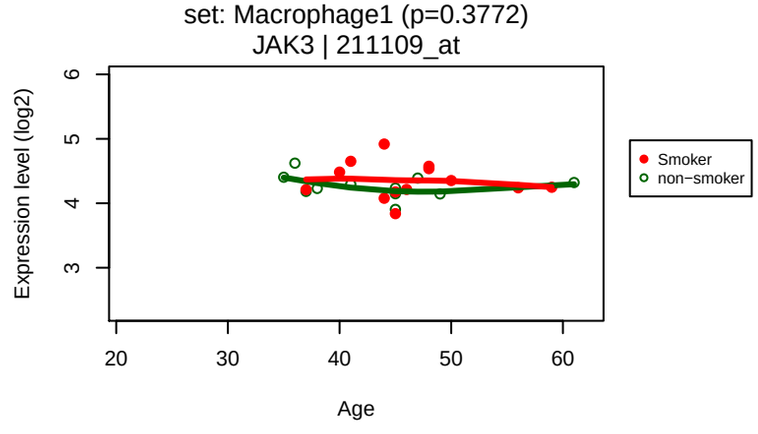

set: SmallAirway1 (p=0.6021)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker

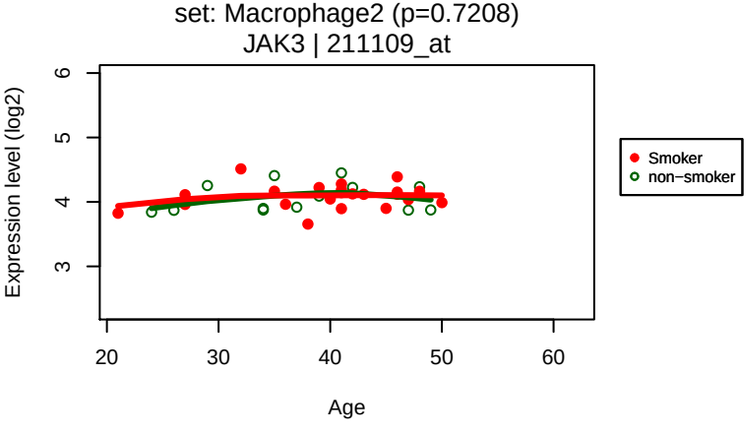

set: SmallAirway2 (p=0.7538)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker

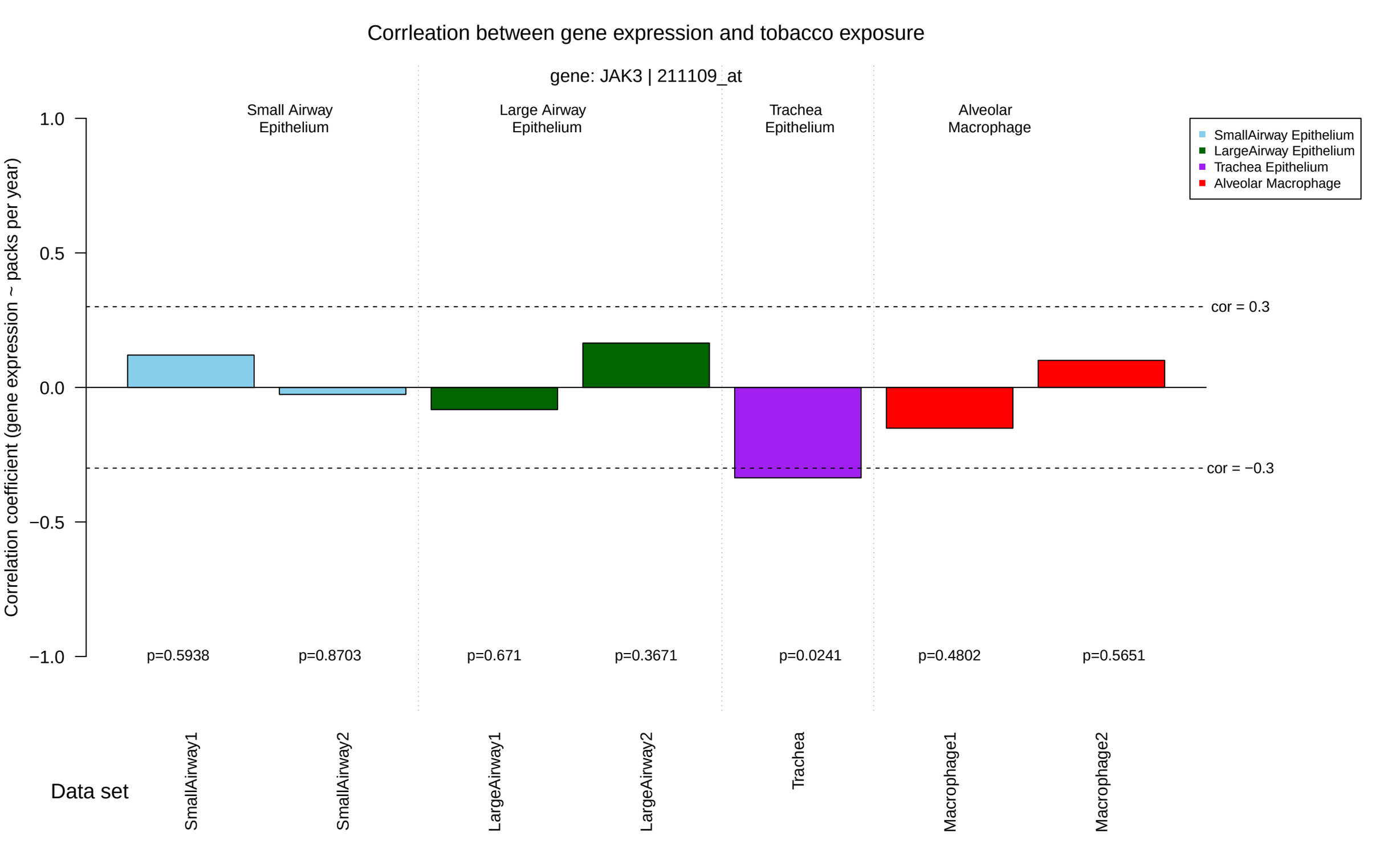

set: LargeAirway1 (p=0.6947)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1883)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1376)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3772)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7208)
JAK3 | 211109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK3 | 211109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5938
p=0.8703
p=0.671
p=0.3671
p=0.0241
p=0.4802
p=0.5651
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2